

Q1 Compute the time complexity using frequency count for the following snippets.

```
a) for(i=1; i<10; i=i+2)    1+6+5
{
    i=i&10;                5
    printf("%d", i);        5
}                            22
```

b)  $i=1;$   $\longrightarrow$  1 unit of time  
 while ( $i \leq 50$ );  
 {  
     Print ("hello");  
 }

c) for ( i=1; i<=20; i++);      1+19+1  $\rightarrow$  21

d)  $\text{for}(i=1; i \leq n; i=i+1) \{$   
 $\quad \text{printf}("hello");$   
 $\quad \text{printf}("world");$   
 $\quad \text{printf}("i = %d", i);$   
 $\}$

$1 + \frac{n}{2} + 1 + \frac{n}{2}$   
 $\frac{n}{2}$   
 $\frac{n}{2}$   
 $\frac{n}{2} \Rightarrow \frac{5n}{2} + 2$